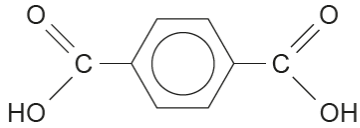
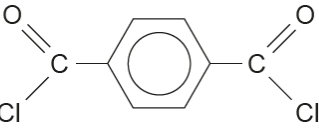
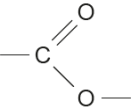


Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		nickel / platinum	1			1		
		(ii)	I	cyclohexanol contains an O—H bond at 3200 to 3550 cm ⁻¹ / cyclohexanol contains a C—O bond at 1000 to 1300 cm ⁻¹ (1) cyclohexanone contains a C=O bond at 1650 to 1750 cm ⁻¹ (1) award (1) for partial answer to both points e.g. cyclohexanone gives peak at 1700 and cyclohexanol gives peak at 3300	1 1			2		
			II	cyclohexanol, C ₆ H ₁₁ OH $M_r = 100.12$ % oxygen = $\frac{16 \times 100}{100.12} = 15.98$ (1) cyclohexanone, C ₆ H ₁₀ O $M_r = 98.10$ % oxygen = $\frac{16 \times 100}{98.10} = 16.31$ (1) these two percentage figures are too close for accurate determination of the proportions present (1)		1 1	1	3		
			III	% by volume = $\frac{49 \times 100}{84} = 58$	1			1		
		(iii)		acidified dichromate / Cr ₂ O ₇ ²⁻ , H ⁺ acidified manganate(VII) / MnO ₄ ⁻ , H ⁺	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			 (1) (1)  or 		2		2		
	(c)	(i)		the polymer is made from an alkene / the monomer has a C=C bond / no additional compound is formed (when polymerisation occurs) (1) award (1) for any of following a polyester contains a  group in the chain there is no ester <u>linkage</u> a polyester is made from an alcohol and a carboxylic acid / acid chloride			2	2		
		(ii)		mass of polymer used = $\frac{150}{300} = 0.5 \text{ g}$ (1) $M_r = \frac{0.50}{4.0 \times 10^{-6}} = 125\,000$ (1)		1		2	1	
				Question 8 total	5	5	4	14	1	1